Summary in Graph

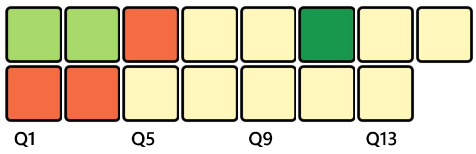
Exam Summary (GO Classes DA Test Series 2026 | Machine Learning | Topic Wise Test 8)

Qs. Attempted:	**6** (5 + 1)	**Correct Marks:**	**4** (2 + 2)
Correct Attempts:	**3** (2 + 1)	**Penalty Marks:**	**1** (1 + 0)
Incorrect Attempts:	**3** (3 + 0)	**Resultant Marks:**	**3** (1 + 2)

Total Questions:	**15** (5 + 10)
Total Marks:	**25** (5 + 20)
Exam Duration:	**45 Minutes**
Time Taken:	**45 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (35) out of 347 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Given a dataset $\{0, 2, 4, 6, 24, 26\}$, initialize the k-means clustering algorithm with 2 cluster centers: $c_1 = 3$ and $c_2 = 4$. After running k-means for two iterations, what are the values of c_1 and C_2 after the first and second iterations?

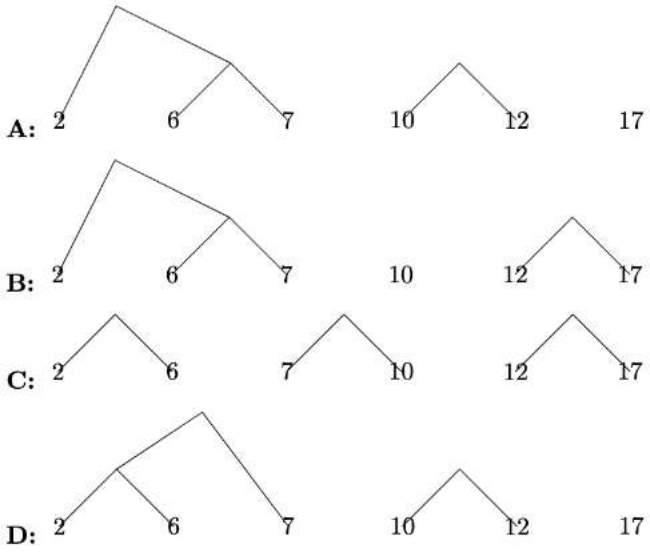
A. After the first iteration: $c_1 = 1, c_2 = 15$; After the second iteration: $c_1 = 3, c_2 = 25$
B. After the first iteration: $c_1 = 2, c_2 = 14$; After the second iteration: $c_1 = 3, c_2 = 25$

C. After the first iteration: $c_1 = 1, c_2 = 15$; After the second iteration: $c_1 = 4, c_2 = 24$
D. After the first iteration: $c_1 = 2, c_2 = 16$; After the second iteration: $c_1 = 3, c_2 = 24$

Your Answer: A | Correct Answer: A | Correct | Time taken: 02min 18sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Run the first three iterations of hierarchical clustering using complete linkage on the 1D dataset $X = \{2, 6, 7, 10, 12, 17\}$. What is the resulting (partial) dendrogram after these iterations?



Your Answer: D | Correct Answer: A | Incorrect | Time taken: 02min 47sec | Discuss

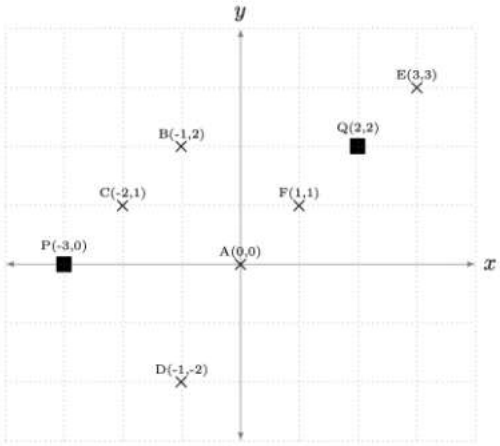
Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider Euclidean distance in $\mathbb{R}$. There are three clusters: $A = (0, 2, 6), B = (3, 9), C = (11)$. Which two clusters will single linkage and complete linkage merge next, respectively?

A. AB and AB
B. AB and AC
C. AB and BC
D. AC and AC

Your Answer: C | Correct Answer: C | Correct | Time taken: 01min 40sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning



In the first step of K-means with the standard Euclidean distance metric, which points will be assigned to the cluster centered at P?

A. C, D
B. A, C, D
C. B, C, D
D. A, B, C

Your Answer: A | Correct Answer: C | Incorrect | Time taken: 02min 13sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Consider the following 1-dimensional dataset, where each point is labeled with a number:
·$A : x_a = 1$
·$B : x_B = 4$
·$C : x_C = 4.5$
·$D : x_D = 6$
·$E : x_E = 10$
Suppose you perform Agglomerative Hierarchical Clustering with single linkage. What is the sequence at which the points will be joined? If two points are at the same level, prioritize joining them in alphabetical order.

A. ABCDE
B. BCDAE
C. BACDE
D. ACDBE

Your Answer: A | Correct Answer: B | Incorrect | Time taken: 03min 41sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Which of the following methods will cluster the data in panel (a) of the figure below into the two clusters (red circle and blue horizontal line) shown in panel (b)? Every dot in the circle and the line is a data point. In all the options that involve hierarchical clustering, the algorithm is run until we obtain two clusters.



A. Hierarchical agglomerative clustering with Euclidean distance and complete linkage
B. Hierarchical agglomerative clustering with Euclidean distance and single linkage
C. Hierarchical agglomerative clustering with Euclidean distance and centriod linkage
D. k-means clustering with k=2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 02min 59sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Given the dataset: $(2,2),(4,4),(5,5),(6,6),(9,9),(0,4),(4,0)$, K-Means clustering is performed with $\mathrm{k}=3$ using Manhattan distance as the distance function. The initial clusters are:
$\cdot \mathrm{C}_1:\{(2,2),(4,4),(6,6)\}$
$\cdot \mathrm{C}_2:\{(0,4),(4,0)\}$
$\cdot \mathrm{C}_3:\{(5,5),(9,9)\}$
What are the new centroids after one iteration of K-Means?

A. $\mathrm{C}_1:(3,3),\mathrm{C}_2:(2,2),\mathrm{C}_3:(7,7)$
B. $\mathrm{C}_1:(4,4),\mathrm{C}_2:(2,2),\mathrm{C}_3:(7,7)$
C. $\mathrm{C}_1:(4,4),\mathrm{C}_2:(3,3),\mathrm{C}_3:(7,7)$
D. $\mathrm{C}_1:(3,3),\mathrm{C}_2:(2,2),\mathrm{C}_3:(6,6)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 01min 06sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

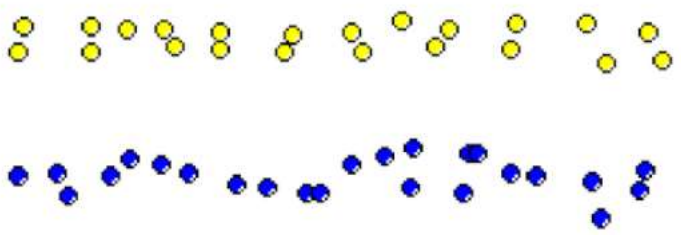
k-means refers to optimizing which of the following objectives? Here $\mu_{C(j)}$ is the mean of the cluster that x_j belongs to. m is the number of points.

A. $F(\mu,C)=\sum_{j=1}^{m}\left\|\mu_{C(j)}-x_j\right\|_2^2$
B. $F(\mu,C)=\min_{j=1}^{m}\left\|\mu_{C(j)}-x_j\right\|_2^2$
C. $F(\mu,C)=\sum_{j=1}^{m}\left\|\mu_{C(j)}-x_j\right\|$
D. $F(\mu,C)=\max_{j=1}^{m}\left\|\mu_{C(j)}-x_j\right\|_2^2$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 21sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Which clustering method is most likely to produce the following results as clusters? Choose the most likely method:



A. Hierarchical clustering with single link
B. Hierarchical clustering with complete link
C. Hierarchical clustering with average link

D. K-means (k=2)

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 01min 05sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose K-Means with $K = 2$ is used to cluster the dataset $\{5, 7, 8, 10, x\}$ with initial cluster centers:

$$c_1 = 5, \quad c_2 = 10$$

What is x if one of the cluster centers in the next iteration is 5 ? Here, x can belong to either cluster 1 or 2 .

A. $x = 3$
B. $x = -3$
C. $x = 7$
D. $x = 10$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 08sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Considering single-link and complete-link hierarchical clustering, is it possible for a point to be closer to points in other clusters than to points in its own cluster? If so, in which approach will this tend to be observed?

A. No
B. Yes, single-link clustering
C. Yes, complete-link clustering
D. Yes, both single-link and complete-link clustering

Your Answer: D | Correct Answer: D | Correct | Time taken: 00min 24sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the following 1-dimensional dataset, where each point is labeled with a number:
$\cdot A : x_a = 1$
$\cdot B : x_B = 4$
$\cdot C : x_C = 4.5$
$\cdot \mathrm{D} : \mathrm{x_D} = 6$
$\cdot E : x_E = 10$
Suppose we perform Agglomerative Hierarchical Clustering with single linkage. If the dendrogram for the hierarchical clustering is cut at a height of 3.5 units, how many clusters will be formed?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 14sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose you have an unlabeled dataset:

$$\{x_1, x_2, \ldots, x_N\}$$

You run k -means with 50 different random initializations (always with the same value of k), and obtain 50 different clusterings of the data. What is the recommended way for choosing which one of the clusterings to

use? (Select one)

A. Always pick the final (50th) clustering found, since by that time it is more likely to have converged to a good solution.
B. The only way to do so is if we also have labels y_i for our data.
C. Choose the clustering with smallest $\frac{1}{N}\sum_{i=1}^{N}\|x_i - \mu_{z_i}\|^2$, where μ_{z_i} is the mean of the cluster to which x_i is assigned.
D. The answer is ambiguous, and there is no good way of choosing.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 04sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the 1D dataset: $x_i = i$ for $i = 1$ to 25 . To select good initial cluster centers for k-means where $k = 5$, we set $c_1 = 1$. Then, subsequent centers c_j are selected from the unused points in the dataset such that each is farthest from any already-selected centers $c_1, \ldots, c_{j-1}$.
Which of the following represents the correct sequence of cluster centers c_1, c_2, c_3, c_4, c_5 ?

A. $c_1 = 1, c_2 = 25, c_3 = 13, c_4 = 7, c_5 = 19$ (arbitrary order for the last two).
B. $c_1 = 1, c_2 = 13, c_3 = 25, c_4 = 19, c_5 = 7$.
C. $c_1 = 1, c_2 = 25, c_3 = 19, c_4 = 7, c_5 = 13$.
D. $c_1 = 1, c_2 = 19, c_3 = 25, c_4 = 7, c_5 = 13$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 03sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

With access to an n-by- n matrix of pairwise data distances, but no access to the data itself, we can use which of the following clustering techniques: A) k-means; B) k-medoids; C) hierarchical clustering:

A. A and B
B. B and C
C. A and C
D. A, B, and C

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 26min 20sec | Discuss

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